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Sample Question Set - Unit 2

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO2

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 2

Unit Title: Data Preparation and Machine Learning Workflow

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the Pandas structure that holds a single labelled column of data.

- A) A Series
- B) A DataFrame
- C) An ndarray
- D) A dictionary

Q2. Identify the Pandas method that returns count, mean, standard deviation and quartiles for the numeric columns.

- A) info()
- B) describe()
- C) head()
- D) shape

Q3. State the Matplotlib plot used to show the distribution of a single numeric column.

- A) A scatter plot
- B) A line plot
- C) A histogram
- D) A pie chart

Q4. Name the kind of dataset made up of photographs and free text.

- A) Structured data

- B) Tabular data
- C) Relational data
- D) Unstructured data

Q5. Identify the scaler that uses the median and the interquartile range rather than the mean and standard deviation.

- A) The standard scaler
- B) The robust scaler
- C) The min-max scaler
- D) The normaliser

Q6. State what a validation partition is used for.

- A) Fitting the parameters of the final model
- B) Reporting the final performance of the system
- C) Storing the records that have missing values
- D) Choosing between models and settings before the test set is used

Q7. Explain why the test partition should be consulted only once.

- A) Repeated use turns it into part of the development data
- B) Repeated use damages the stored data file
- C) Repeated use makes the model train more slowly
- D) Repeated use changes the model's parameters

Q8. Distinguish a Pandas DataFrame from a NumPy array.

- A) A NumPy array carries column names and mixed types
- B) Both of the two carry column names and mixed types
- C) A DataFrame carries column names and mixed types
- D) Neither of the two can hold numeric values

Q9. Describe what a correlation close to zero between two features indicates.

- A) No relationship of any kind between the two
- B) The two features are identical to each other
- C) No straight-line relationship between the two
- D) One feature causes the other to change

Q10. Interpret a column in which 70 per cent of the entries are missing.

- A) The column is usually dropped rather than imputed
- B) The column should be filled with its mean value
- C) The rows with a value should all be deleted
- D) The column is the most informative in the table

Q11. Predict the effect of applying one-hot encoding to a column holding five thousand distinct categories.

- A) The column is reduced to a single column
- B) The categories are automatically merged
- C) The encoding fails and returns an error
- D) Five thousand sparse columns are created

Q12. Recognize the plot described: vertical bars showing how many records fall into each category.

- A) A histogram
- B) A bar chart
- C) A scatter plot
- D) A box plot

Q13. Explain why a tree-based model does not need its features placed on a common scale.

- A) It converts every column to the same units
- B) It ignores every column except the largest
- C) It computes distances between the records
- D) It splits on thresholds within a single column

Q14. Summarize what exploratory data analysis is carried out for.

- A) To fit the final model to the training data
- B) To report the accuracy of the finished system
- C) To learn what is in the data before any model is chosen

- D) To deploy the model into the live product

Q15. Distinguish structured data from semi-structured data.

- A) Semi-structured data has fixed columns and no tags
- B) Structured data has fixed columns, semi-structured carries its own tags
- C) Both of the two have a fixed set of columns
- D) Neither of the two can be stored in a file

Q16. Explain why an indicator column is sometimes added when missing values are filled in.

- A) The fact that a value was absent may itself be informative
- B) The indicator improves the speed of training
- C) The indicator is required by the Pandas library
- D) The indicator replaces the need for imputation

Q17. Predict what happens to the number of rows when duplicate records are removed.

- A) It rises by the number of duplicates found
- B) It falls by the number of duplicates removed
- C) It stays exactly the same as before
- D) It falls to the number of distinct columns

Q18. Describe what stratified k-fold cross validation preserves.

- A) The proportion of each class within every fold
- B) The original order of the rows in the table
- C) The exact number of features in every fold
- D) The mean of every numeric column per fold

Q19. Explain why a logarithmic transformation is applied to a heavily right-skewed column.

- A) It removes every missing value from the column
- B) It converts the column into a categorical one
- C) It guarantees the column becomes normal
- D) It compresses the long tail and evens out the spread

Q20. Distinguish filter feature selection from wrapper feature selection.

- A) A wrapper scores features alone, a filter tests them
- B) Both of the two test the features inside a model
- C) A filter scores features alone, a wrapper tests them in a model
- D) Both of the two score the features in isolation

Q21. Apply 10-fold cross validation to a dataset of 1,500 rows and determine the number of rows in the validation partition of each fold.

- A) 150

- B) 1,350
- C) 15
- D) 1,500

Q22. Solve the following: a dataset of 2,000 rows is split with a test size of 0.2. Determine the number of training rows.

- A) 400
- B) 1,800
- C) 1,600
- D) 200

Q23. Use the interquartile range rule with a multiplier of 1.5 on a column whose first quartile is 20 and third quartile is 60, and determine the upper fence.

- A) 100
- B) 120
- C) 90
- D) 140

Q24. Compute the standardised value of an observation equal to 88 in a column whose mean is 70 and standard deviation is 12.

- A) -1.5
- B) 0.75
- C) 2.0
- D) 1.5

Q25. Apply min-max scaling to the value 45 in a column whose minimum is 5 and maximum is 105.

- A) 0.45
- B) 0.5
- C) 0.4
- D) 0.35

Q26. Examine the following and determine the fault: a categorical column holding the cities Delhi, Mumbai and Chennai is encoded as 0, 1 and 2 and fed to a linear model.

- A) The encoding produces too many extra columns
- B) The encoding removes the missing values silently
- C) The encoding requires the column to be scaled
- D) An order and a spacing have been imposed that do not exist

Q27. Solve the following: a column of 800 entries has 120 missing values. Determine the percentage that is missing.

- A) 15 per cent
- B) 12 per cent
- C) 20 per cent

- D) 8 per cent

Q28. Apply the correct order of steps when a numeric column must be scaled before a model is trained.

- A) Fit the scaler on all rows, then split into two parts
- B) Split first, fit the scaler on the training rows, transform both
- C) Split first, fit a separate scaler on each partition
- D) Transform both partitions before any scaler is fitted

Q29. Use Pandas to determine the method that replaces missing entries with a stated value.

- A) dropna()
- B) isnull()
- C) describe()
- D) fillna()

Q30. Compute the number of columns produced by one-hot encoding a four-category column when no category is dropped.

- A) 3
- B) 4
- C) 1
- D) 16

Q31. Apply 5-fold cross validation and determine how many times the model is fitted.

- A) 1
- B) 10
- C) 5
- D) 25

Q32. Examine a scatter plot of two numeric columns showing a clear upward straight-line trend, and determine what it indicates.

- A) A strong positive linear relationship
- B) A strong negative linear relationship
- C) No relationship between the columns
- D) A curved relationship between them

Q33. Use the shape attribute on a DataFrame holding 500 rows and 12 columns and determine what it returns.

- A) (500, 12)
- B) (12, 500)
- C) 500
- D) 512

Q34. Solve the following: a table of 1,000 rows has 45 duplicates and 30 rows with a missing target removed, with no overlap between the two groups. Determine the rows that remain.

- A) 955
- B) 970
- C) 875
- D) 925

- C) The size of the test set does not affect the estimate
- D) A smaller test set always gives a better estimate

----- End of Sample Question Set -----

Q35. Apply feature engineering to a column of order timestamps and determine a useful derived feature.

- A) The raw timestamp converted to a text string
- B) The hour of the day and the day of the week
- C) The row number at which the order appears
- D) The number of characters in the timestamp

Q36. Compare a histogram with a box plot on what each one reveals about a single numeric column.

- A) The box plot shows the shape, the histogram the quartiles
- B) Both of the two show only the mean of the column
- C) The histogram shows the shape, the box plot the quartiles and outliers
- D) Both of the two show the relationship between columns

Q37. Use a variance threshold for feature selection and determine which feature it removes.

- A) The one with the largest range of values
- B) The one most strongly correlated with the target
- C) The one whose value is nearly the same in every row
- D) The one with the greatest number of categories

Q38. Examine a training set in which one class holds only 2 per cent of the rows and determine the appropriate splitting strategy.

- A) A purely random split
- B) A stratified split
- C) A split ordered by the target
- D) No split, using all rows for training

Q39. Apply the train, validation and test arrangement and determine which partition is used to choose the settings.

- A) The training partition
- B) The test partition
- C) All three of the partitions
- D) The validation partition

Q40. Relate the size of the test partition to the reliability of the estimate it produces.

- A) A larger test set gives a steadier estimate but less training data
- B) A larger test set gives a less steady estimate